Pedigree analysis and relatedness inference

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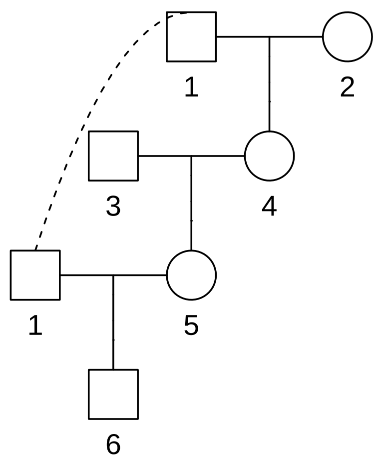
Solutions for exercise set I: Pedigrees and measures of relatedness

Many of these exercises can be solved with QuickPed (https://magnusdv.shinyapps.io/quickped). Some alternative solutions using R are included below.

Solution I-1

a) Solution in R:

```
x = linearPed(2, sex = 2) |> addSon(parents = c(1, 5))
plot(x)
```



b) $f = \frac{1}{8} = 0.125$.

c) The inbreeding coefficient of an individual is equal to the kinship coefficient between its parents. In this case, the child's parents are grandfather–granddaughter, with kinship coefficient $\varphi = 1/8$.

Solution I-2

a) Maternal half siblings, and also first cousins through the fathers. The children are not inbred.

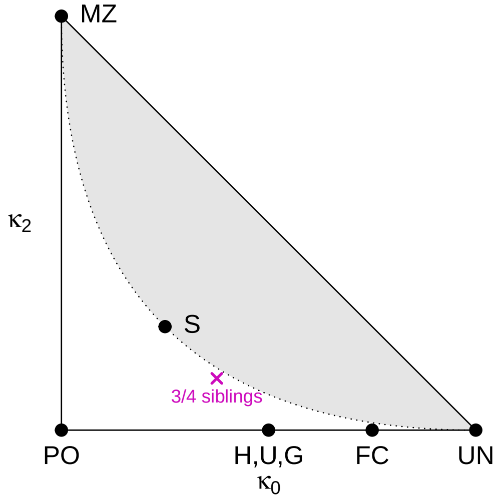
b) Solution in R:

```
x = nuclearPed(2) |> addSon(c(3,5)) |> addSon(c(4,5))
kappaIBD(x, 6:7)
```

```
## [1] 0.375 0.500 0.125
```

c) The point is shown in the triangle below. R code:

```
showInTriangle(kappa = c(3/8, 1/2, 1/8), label = "3/4 siblings")
```



d) The point is exactly halfway between half and full siblings. (And $\frac{3}{4}$ is halfway between $\frac{1}{2}$ and 1.)

Solution I-3

a) (Omitted.)

b) 6 is both an uncle and half-uncle of 7.

c) Their kinship coefficient is $3/16 = 0.1875$.

d) The inbreeding coefficient of 7 is $1/8 = 0.125$.

e) (Omitted)

Solution I-4

a) The kinship coefficient φ between A and B is the probability that a random allele from A is IBD to a random allele from B at the same locus.

b) Outbred monozygotic twins have kinship coefficient $\varphi = 0.5$.

c) This requires inbreeding. For example, 4 generations of full-sib mating gives $\varphi \approx 59\%$.

d) To achieve $\varphi = 1$ the two individuals must be 100% inbred and clones of each other. In a standard pedigree this is possible only asymptotically, e.g., after infinitely many generations of full-sib mating.

Solution I-5

a) (Omitted.)

b) They typically share around 25 segments (range ~ 20-30), of average length varying from 40 cM to 80 cM.

c) The "Total length IBD" distribution peaks around 50%, in agreement with the theoretical $\kappa_1 = 0.5$.

d) The female version has markedly more segments (typically around 30-35), but shorter average length (rarely above 60 cM). The difference is caused by the sex of the middle person: Males on average recombine less, resulting in fewer/longer segments.

e) The distributions should be very similar, since the sex of the grandparent does not matter! Regardless of recombination in the grandparental meiosis, the middle person always receives an entire copy of each chromosome.